

The Salt Problem—Making a Mountain Out of Molehills

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Summary

Given the safety-height recommendation of the panel, we find the maximum height for the salt pile should be approximately 21.7 ft.

We developed a model employing a loading strategy to obtain a function of three parameters for maximum height. We determined that the function is stable around the solution. Variation of 10% among the three parameters produced only an 8% variation in the solution.

We implemented a computer model to generate three-dimensional images of the salt pile, to calculate volumes, and to reflect our loading strategy.

Summary

A Midwestern state must store salt for the winter. Historically, the salt has been stored in circular domed buildings using front-end loaders. The salt has been amassed to heights of 25–30 ft. Recently, a panel determined that it was unsafe to pile salt above 15 ft using front-end loaders.

We interpret the problem as finding a method to maximize the amount of salt stored (consequently its piled height) while maintaining the “absolute” safety recommendation of the panel.

Assumptions

- The roof of the dome will not support any side load.
- The retaining walls will support any 4-ft side load.
- The height of the pile of salt is measured from the floor of the dome.

- The panel's recommendation is an "absolute safe" limit that allows the front-end loader to traverse any point on the pile's surface without sliding.

Definitions

The *critical* (or *slip*) *angle* of a noncohesive granular substance (salt) is the maximum angle, as measured from the horizontal, that the substance will slide when piled. For example, when a mass is placed near a slope with critical angle, it causes the material to flow, taking the mass with it, until a new critical angle is achieved. This is the *critical angle under load* and is dependent on the mass. Therefore, there are two notable angles.

For a front-end loader, there is a *radius of reach* for the bucket.

Analysis

One type of analysis, the micro view, is to model the physical parameters of the salt, the mass of the loader, and how the salt behaves with vibration and varying forces of shear and stress. Alternatively, the macro view, while likely to be less rigorous, will be shown to depend on only three parameters:

- the slip angle (in degrees) of salt, which we denote by α ;
- the slip angle (in degrees) of the front-end loader on salt, which we denote by β ;
- the radius of reach of the front-end loader, which we denote by r .

We feel that these three parameters can either either derived or—better yet—determined experimentally. Further, this simplification allows us to concentrate on the method for piling salt to achieve the maximum height while retaining the panel's recommendation for "absolute" safety.

Design of the Model

First, we analyze the front-end loader. We impose a coordinate that we have called the radius of reach, which defines the distance that the loader's bucket can reach while dumping. This coordinate is measured from the base of the wheels.

The loader can then make piles of salt determined by the critical angle of the salt and the loader's radius of reach (see **Figure 1**). The height of the salt piles made is then

$$h_L = r \sin \alpha.$$

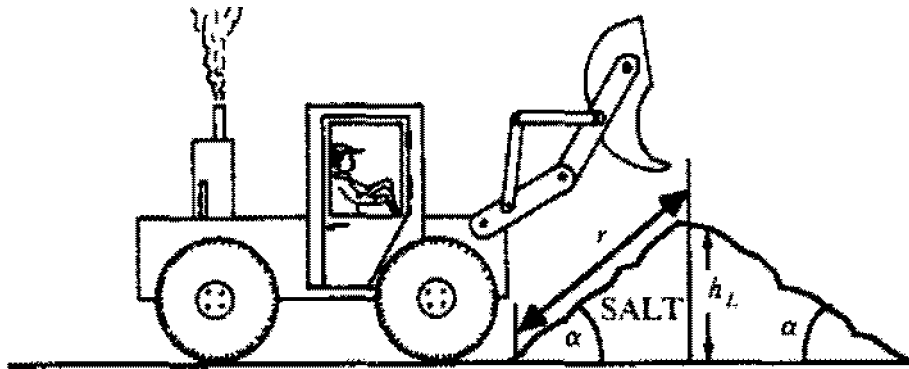


Figure 1. The front-end loader on a horizontal surface.

When the loader is on a positive slope (see **Figure 2**), the height h_{AU} of a salt pile is determined by

$$\frac{\sin(90^\circ + \beta)}{r} = \frac{\sin(\alpha - \beta)}{h_{AU}},$$

so that

$$h_{AU} = \frac{r \sin(\alpha - \beta)}{\cos \beta}.$$

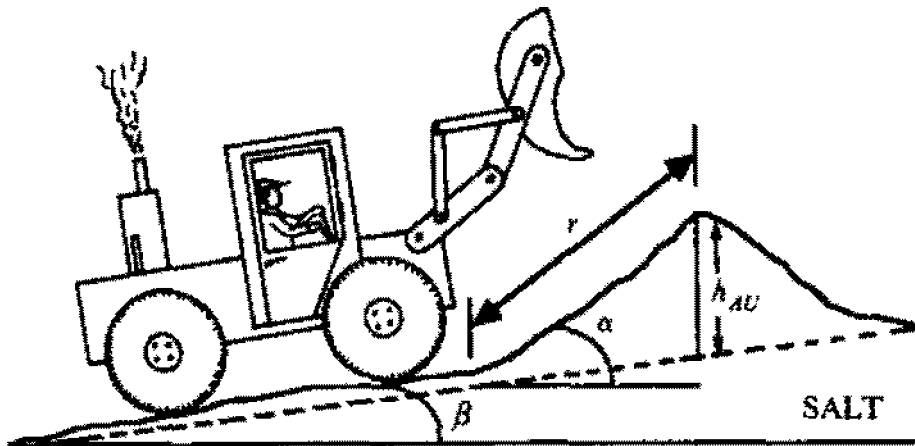


Figure 2. The front-end loader on a positive slope.

Conversely, if the loader is on a negative slope (see **Figure 3**), the height h_{AD} of a salt pile is determined by

$$\frac{\sin(90^\circ - \beta)}{r} = \frac{\sin(\alpha + \beta)}{h_{AD}},$$

so that

$$h_{AD} = \frac{r \sin(\alpha + \beta)}{\cos \beta}.$$

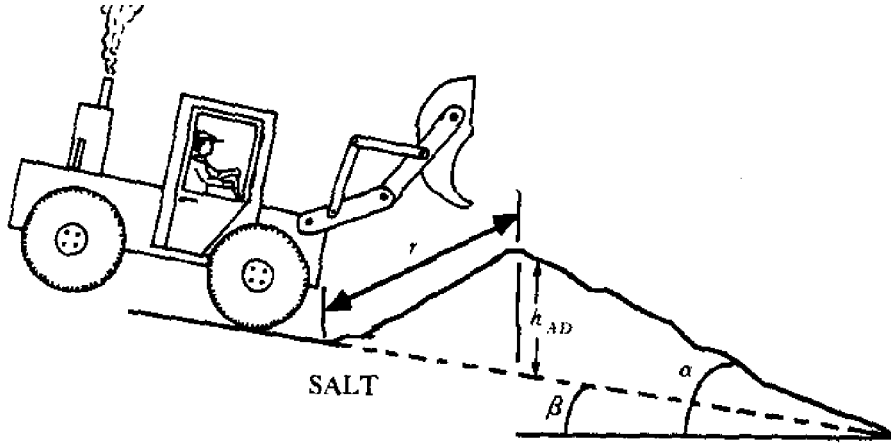


Figure 3. The front-end loader on a negative slope.

Now that we understand the limits of the loader, we next analyze the salt pile within the storage dome. The “absolute” safe condition is shown in **Figure 4**, where β is the critical angle of the loader on salt. By our assumption, the loader may go anywhere upon the conical surface. The angle β is given by

$$\beta = \arctan\left(\frac{15 - 4}{51.5}\right) = 12.06^\circ.$$

Further, since heights of 25–30 ft are possible, and assuming a value for h_{AU} of approximately 5 ft (see **Figure 5**), we have

$$\alpha = \arctan\left[\frac{25 - 4}{100 - \frac{25 - 4 - 5}{\tan 12.06^\circ}}\right] = 39.9^\circ.$$

This value leads to a radius of reach for the loader of

$$r = \frac{h_{AU} \cos \beta}{\sin(\alpha - \beta)} = 10.5 \text{ ft.}$$

Since r is 10.5 ft, it is reasonable to expect that h_{AU} is approximately 5 ft, or $r/2$. These values are the ones that we use to determine the maximum height of salt.

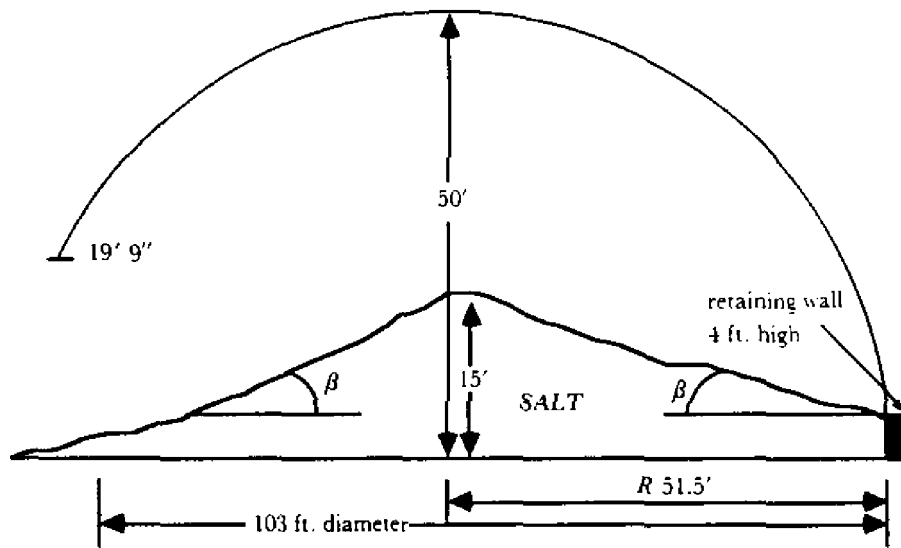


Figure 4. The salt pile.

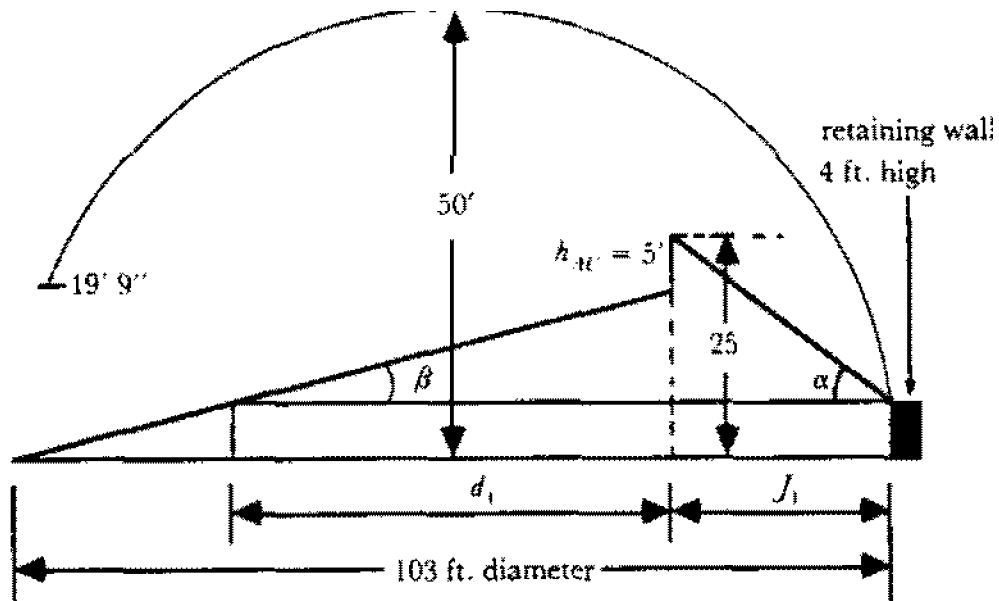


Figure 5. Determining the values of the parameters.

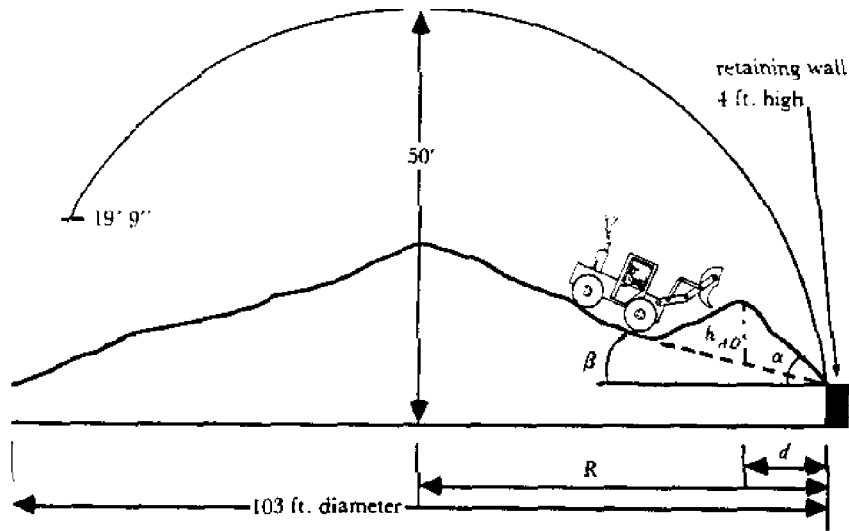


Figure 6. The panel's recommendation for the salt pile.

We believe that **Figure 6** represents the panel's recommendation and is a safe surface.

Additional storage is possible if the the loading operation follows the following procedure:

- Fill the dome until the "absolute" safe condition is met.
- Add salt on top of the safe surface, starting opposite the door and working around the periphery of the dome. The driver is instructed never to drive on the added piles.
- New deposits are positioned against the previous loads.
- To ensure stability, the loader should drive along the gradient. To avoid turns on the incline, all turns may be made at the top of the safe surface.
- Finally, the loader backfills the route up the incline to the door.

If this simple procedure is followed, more salt may be stored in the dome while maintaining "absolute" safety. The maximum height (in ft) of the salt in a dome of radius R is then

$$\begin{aligned} h_{\max} &= 4 + \frac{\tan \beta [2R - r \sin(90^\circ - \alpha) \cos \beta]}{r \sin(90^\circ - \beta)} + \frac{r \sin(\alpha + \beta)}{\sin(90^\circ - \beta)} \\ &= 4 + 2R \tan \beta + r \sin \alpha. \end{aligned}$$

Using the values obtained before, we get

$$h_{\max} = 21.7 \text{ ft.}$$

The final cross-section of the salt mound is illustrated in **Figure 7**.

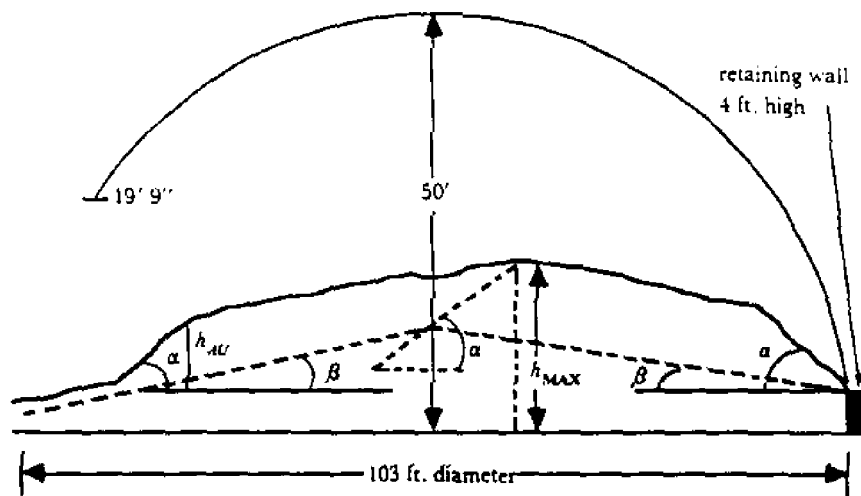


Figure 7. Final cross-section of the mound.

Testing and Evaluation of the Model

Testing of a mathematical model implies some method to check its validity. Since safety is a prime concern, we recommend that a realistic test be conducted. This would involve building a pile of salt in an open area, where the front-end loader could slide freely. With the values that are experimentally obtained, we can use the equations above to generate a three-dimensional surface (see **Figure 8**). Moreover, this surface can be used for demonstration and instruction of the salt-loading technique that we outlined.

To approximate capacity of the salt pile, we applied Simpson's rule to integrate the surface numerically. For the values in our example, the volume is 3,892 cu yd.

Strengths and Weaknesses

The strength of this model is its simplicity. If we had to explain to a layperson how we arrived at our recommendation, we could illustrate it without difficulty.

There are three specific parameters required for the use of the model, and these may be identified easily.

The weakness of this model is also its simplicity. In using only macro parameters, we may have overlooked a crucial point.

Our sensitivity analysis yields

$$\Delta h_{\max} = (\sin \alpha) \Delta r + (r \cos \alpha) \Delta \alpha + (2R \sec^2 \beta) \Delta \beta.$$

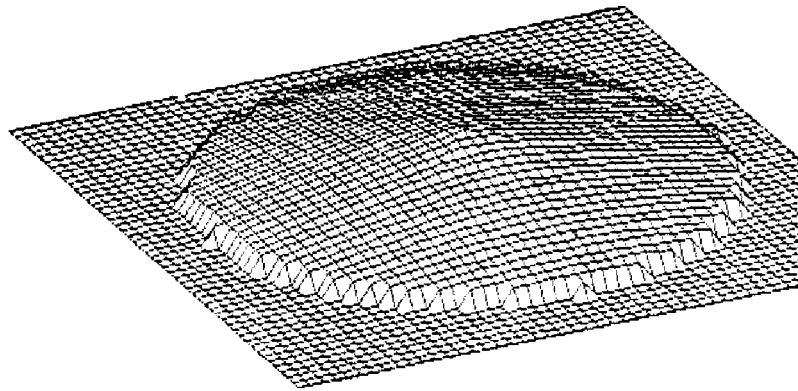


Figure 8a. Safe surface.

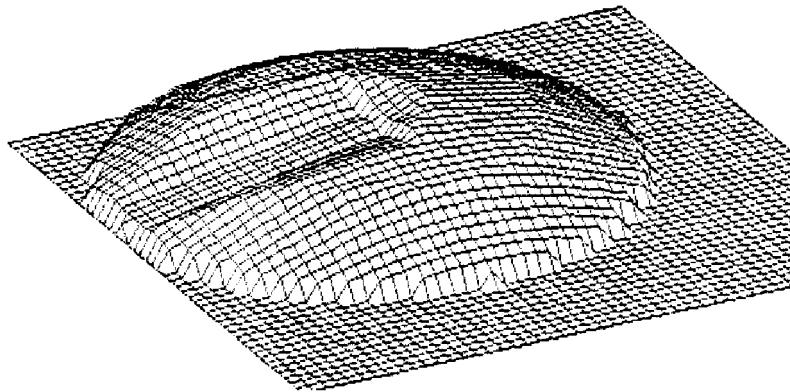


Figure 8b. Second layer.

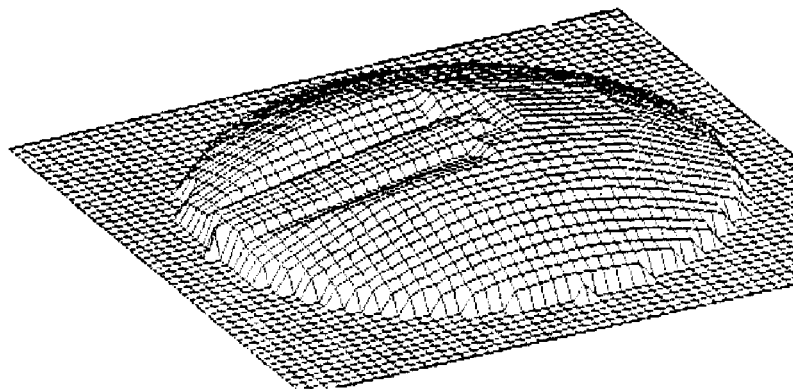


Figure 8c. Back fill.

Using the derived values for α , β , and r , and letting each of them vary by 10%, then

$$\Delta h_{\max} = \pm 1.8 \text{ ft},$$

or approximately 8%. Given these values, it appears that our method is stable.

References

- Scott, Ronald F., and Jack J. Schoustra. 1968. *Soil: Mechanics and Engineering*. New York: McGraw-Hill.
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- Taylor, A.B. 1986. *Mathematical Models in Applied Mechanics*. Oxford: Clarendon Press.

Acknowledgment

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